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Roll No

EE-7201-GS
B.E. VII Semester
Examination, December 2020
Grading System (GS)
High Voltage Engineering
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. Explain the tesla coil. How is damped high frequency oscillations obtained from a tesla coil?
2. Explain the partial discharge test on high voltage cables. How is a fault in the insulation located in this test.
3.
 - a) Derive an expression for ripple voltage of a multistage Cockcroft-Walton circuit.
 - b) Explain clearly basic principle of operation of an electrostatic generator.
4.
 - a) What are the requirement of a sphere gap for measurement of high voltage? Discuss the disadvantages of sphere gap for measurements.
 - b) Explain the principle of operation of electrostatic volt meter.
5. Discuss the terms “mobility of ions and electrons”, “diffusion coefficient” and “electron energy distributions”.

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6. Explain the generating voltmeter used for measuring high dc voltages. How does it compare with a potential divider for measuring high dc voltages?
7. Explain the various theories that explain breakdown in commercial liquid dielectrics.
8. Explain the following (Any two) :
 - a) Need for generating high voltages in laboratory.
 - b) Important applications of high voltage.
 - c) BIL.
 - d) Methods of insulation co-ordinator.
